Lab 4 FDSAI

2026-2027

characteristic population

1. Salaries of entry-level computer engineers have a normal distribution with unknown mean and variance. 12 randomly selected computer engineers have monthly salaries (in $): 1074.67, 1011.17, 1089.77, 1177.30, 1001.58, 1001.59, 1182.92, 1101.74, 978.05, 1079.26, 978.66, 978.43.

   mean is the parameter

   a. Construct a 90% confidence interval for the average salary of an entry-level computer engineer.
   b. Looking at this sample, one may think that the starting salaries have a great deal of variability. Construct a 90% confidence interval for the standard deviation of entry-level salaries.
   c. A new random sample of 100 entry-level computer engineers gives the same sample mean and sample standard deviation for the salary as the smaller sample above. Construct 90% confidence intervals for the average salary and for the standard deviation of the salary. Comment on the results.

2. In this exercise, we will reuse the NHANES dataset from Lab 1. The variables BMXHT and BMXWT, which we will examine, refer to the height (cm) and body weight (kg) of surveyed subjects. We will also consider their gender, recorded in the binary variable RIAGENDR (coded as: 1 = male, 2 = female).
   a. (Lab 1) Extract only the information about gender, height, and body weight from the dataset. Check if the new dataset contains missing entries (NaN) and remove them if necessary.
   b. Determine a 95% confidence interval for the mean height and body weight, both separately by gender and for the entire sample. Repeat for a 99% confidence level. What do you observe?
   c. Now assume the entire statistical population is represented by the NHANES dataset. Randomly extract 20 samples of size 100 from it. For each sample, determine a 95% confidence interval for the mean height and mean body weight, both separately by gender and for the entire sample. Plot the resulting confidence intervals within the same graph and highlight the true mean value of the characteristic obtained in the previous step for the entire population. Comment on the results.